

Entities

Create a console project named “entities” with following:

A. Library files:

- entityable.dart with code:

```
abstract class Entityable{  
  String get id;  
  
}
```

- entity.dart contains an abstract class **Entity** implementing to **Entityable**.

```
abstract class Entity implements Entityable {  
  @override  
  String get id;  
}
```

- repository.dart contains an abstract generic class **Repository**

```
abstract class Repository<T extends Entity> {  
  void add(T item);  
  T? byId(String id);  
  Iterable<T> all();  
  bool removeById(String id);  
}
```

- init_mixin.dart contains following code:

```
mixin Initialization<T extends Entity> on Repository<T> {  
  void initialize(List<T> items) {  
    for (var item in items) {  
      add(item);  
    }  
  }  
}
```

- memory_repo.dart contains a generic class **MemoryRepository**

```
class MemoryRepository<T extends Entity> extends Repository<T>  
with Initialization<T>{  
  final Map<String, T> _items = <String, T>{};  
  
  @override  
  
  void add(T item) => _items[item.id] = item;  
  
  @override  
  Iterable<T> all() => _items.values;
```

```

    @override
    T? byId(String id) => _items[id];

    @override
    bool removeById(String id) => _items.remove(id) != null;
}

```

- person_mixin.dart contains following code:

```

mixin PersonMixin {
  String _ssn = '';
  String _name = '';
  int _age = 0;

  String get ssn => _ssn;
  set ssn(String value) {
    if (value.isEmpty) return;
    _ssn = value;
  }

  String get name => _name;
  set name(String value) {
    if (value.isEmpty) return;
    _name = value;
  }

  int get age => _age;
  set age(int value) {
    if (value < 0 || value > 150) return;
    _age = value;
  }
}

```

- person.dart contains a class Person

```

class Person extends Entity with PersonMixin {
  String get info => 'ssn: $ssn, $name, age: $age';

  @override
  String toString() => info;

  @override
  String get id => ssn;

  Person(String ssn, String name, int age) {
    this.ssn = ssn;
    this.name = name;
  }
}

```

```

        this.age = age;
    }
}

```

- student.dart contains a class **Student**

```

class Student extends Person {
    String _university = '';

    String get university => _university;
    set university(String value) {
        if (value.isEmpty) return;
        _university = value;
    }

    @override
    String get info => super.info + ', university: $university';

    Student(String ssn, String name, int age, String university)
        : super(ssn, name, age) {
        this.university = university;
    }
}

```

- employee.dart contains a class **Empolyee**

```

class Employee extends Person {
    String _position = '';
    double _salary = 0.0;

    String get position => _position;
    set position(String value) {
        if (value.isEmpty) return;
        _position = value;
    }

    double get salary => _salary;
    set salary(double value) {
        if (value < 0) return;
        _salary = value;
    }

    @override
    String get info => super.info + ', position: $position, salary: $salary';
}

```

```

        Employee(String ssn, String name, int age, String position, double
salary)
        : super(ssn, name, age) {
            this.position = position;
            this.salary = salary;
        }
    }
}

```

B. Function main()

```

void main(List<String> arguments) {
    MemoryRepository<Person> repo = MemoryRepository<Person>();
    repo.initalize([
        Person('123-45-6789', 'Alice', 30),
        Person('987-65-4321', 'Bob', 25),
        Person('555-55-5555', 'Charlie', 28),
    ]);
    repo.removeById('987-65-4321');
    repo.add(Student('111-22-3333', 'David', 22, 'RUPP University'));
    repo.add(Employee('444-55-6666', 'Eve', 35, 'Tech Corp', 3000.0));

    var targetId = '111-22-3333';
    var person = repo.byId(targetId);
    if (person != null) {
        print('Found person with ID $targetId: $person');
    } else {
        print('Person with ID $targetId not found.');
```

Result:

```

Found person with ID 111-22-3333: ssn: 111-22-3333, David, age: 22, university: RUPP University
ssn: 123-45-6789, Alice, age: 30
ssn: 555-55-5555, Charlie, age: 28
ssn: 111-22-3333, David, age: 22, university: RUPP University
ssn: 444-55-6666, Eve, age: 35, position: Tech Corp, salary: 3000.0)

```